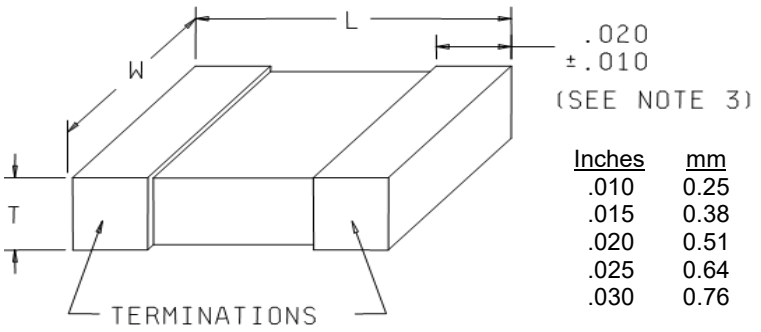
INCH-POUND
MIL-PRF-55681/1G
24 March 2025
SUPERSEDING
MIL-PRF-55681/1F
11 October 2006

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, MULTIPLE LAYER, FIXED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY AND NON-ESTABLISHED RELIABILITY, STYLES CDR01, CDR02, CDR03, AND CDR04

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55681.



Dimensions				
Style	L ± .015 (0.38)	W ± .015 (0.38)	T	
			Min	Max
CDR01	.080 (2.03)	.050 (1.27)	.022 (0.56)	.055 (1.40)
CDR02	.180 (4.57)	.050 (1.27)	.022 (0.56)	.055 (1.40)
CDR03	.180 (4.57)	.080 (2.03)	.022 (0.56)	.080 (2.03)
CDR04	.180 (4.57)	.125 (3.18)	.022 (0.56)	.080 (2.03)

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents, in millimeters, are given for general information only.
3. For style CDR01, termination width shall be .010 inch (0.25 mm), minimum; uncovered band width shall be .030 inch (0.76 mm), minimum.
4. Dimensions and tolerances are for bare chips. For solder coated terminations, for style CDR01 add .020 inch (0.51 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance. For solder coated terminations for styles CDR02, CDR03, and CDR04, add .025 inch (0.64 mm) to the positive length tolerance and .015 inch (0.38 mm) to the positive width and thickness tolerance.

FIGURE 1. Styles CDR01, CDR02, CDR03, and CDR04 capacitors - continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Capacitance value: See table I.

Capacitance tolerance: J = ± 5 percent; K = ± 10 percent; M = ± 20 percent. See table I.

Rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Product level designator: Non-established Reliability - C; Established Reliability - M, P, R, and S.

Marking: In accordance with MIL-PRF-55681.

TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitors characteristics.

Part or Identifying Number (PIN) 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage V_{dc}
CDR01BP100B - - -	10	J, K	BP	100
CDR01BP120BJ - -	12	J	BP	100
CDR01BP150B - - -	15	J, K	BP	100
CDR01BP180BJ - -	18	J	BP	100
CDR01BP220B - - -	22	J, K	BP	100
CDR01BP270BJ - -	27	J	BP	100
CDR01BP330B - - -	33	J, K	BP	100
CDR01BP390BJ - -	39	J	BP	100
CDR01BP470B - - -	47	J, K	BP	100
CDR01BP560BJ - -	56	J	BP	100
CDR01BP680B - - -	68	J, K	BP	100
CDR01BP820BJ - -	82	J	BP	100
CDR01BP101B - - -	100	J, K	BP	100
CDR01B-121B - - -	120	J, K	BP, BX	100
CDR01B-151B - - -	150	J, K	BP, BX	100
CDR01B-181B - - -	180	J, K	BP, BX	100
CDR01BX221B - - -	220	K, M	BX	100
CDR01BX271BK - -	270	K	BX	100
CDR01BX331B - - -	330	K, M	BX	100
CDR01BX391BK - -	390	K	BX	100

See footnote at end of table.

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TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitors characteristics - continued.

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage V_{dc}
CDR01BX471B - - -	470	K, M	BX	100
CDR01BX561BK - -	560	K	BX	100
CDR01BX681B - - -	680	K, M	BX	100
CDR01BX821BK - -	820	K	BX	100
CDR01BX102B - - -	1,000	K, M	BX	100
CDR01BX122BK - -	1,200	K	BX	100
CDR01BX152B - - -	1,500	K, M	BX	100
CDR01BX182BK - -	1,800	K	BX	100
CDR01BX222B - - -	2,200	K, M	BX	100
CDR01BX272BK - -	2,700	K	BX	100
CDR01BX332B - - -	3,300	K, M	BX	100
CDR01BX392AK - -	3,900	K	BX	50
CDR01BX472A - - -	4,700	K, M	BX	50
CDR02BP221B - - -	220	J, K	BP	100
CDR02BP271BJ - -	270	J	BP	100
CDR02BX392BK - -	3,900	K	BX	100
CDR02BX472B - - -	4,700	K, M	BX	100
CDR02BX562BK - -	5,600	K	BX	100
CDR02BX682B - - -	6,800	K, M	BX	100
CDR02BX822BK - -	8,200	K	BX	100
CDR02BX103B - - -	10,000	K, M	BX	100
CDR02BX123AK - -	12,000	K	BX	50
CDR02BX153A - - -	15,000	K, M	BX	50
CDR02BX183AK - -	18,000	K	BX	50
CDR02BX223A - - -	22,000	K, M	BX	50
CDR03BP331B - - -	330	J, K	BP	100
CDR03BP391BJ - -	390	J	BP	100
CDR03BP471B - - -	470	J, K	BP	100
CDR03BP561BJ - -	560	J	BP	100
CDR03BP681B - - -	680	J, K	BP	100
CDR03BP821BJ - -	820	J	BP	100
CDR03BP102B - - -	1,000	J, K	BP	100
CDR03BX123BK - -	12,000	K	BX	100
CDR03BX153B - - -	15,000	K, M	BX	100
CDR03BX183BK - -	18,000	K	BX	100
CDR03BX223B - - -	22,000	K, M	BX	100
CDR03BX273BK - -	27,000	K	BX	100
CDR03BX333B - - -	33,000	K, M	BX	100
CDR03BX393AK - -	39,000	K	BX	50
CDR03BX473A - - -	47,000	K, M	BX	50
CDR03BX563AK - -	56,000	K	BX	50
CDR03BX683A - - -	68,000	K, M	BX	50
CDR04BP122BJ - -	1,200	J	BP	100
CDR04BP152B - - -	1,500	J, K	BP	100
CDR04BP182BJ - -	1,800	J	BP	100
CDR04BP222B - - -	2,200	J, K	BP	100
CDR04BP272BJ - -	2,700	J	BP	100
CDR04BP332B - - -	3,300	J, K	BP	100
CDR04BX393BK - -	39,000	K	BX	100
CDR04BX473B - - -	47,000	K, M	BX	100

See footnote at end of table.

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TABLE I. CDR01, CDR02, CDR03, and CDR04 capacitors characteristics - continued.

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage V_{dc}
CDR04BX563BK - -	56,000	K	BX	100
CDR04BX823AK - -	82,000	K	BX	50
CDR04BX104A - - -	100,000	K, M	BX	50
CDR04BX124AK - -	120,000	K	BX	50
CDR04BX154A - - -	150,000	K, M	BX	50
CDR04BX184AK - -	180,000	K	BX	50

1/ Complete PIN shall include additional symbol(s) to indicate voltage-temperature limit (where applicable), capacitance tolerance (where applicable), termination finish, and product level.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5910-2025-009)

Review activities:
Army - MI
Navy - AS, MC
Air Force - 19

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